Std8 **POLYGONS**

Chapter4 **Practice problems**

1) Given the number of sides. Find the sum of the angles of the polygons

Number of sides	Sum of angles
16	
22	
36	
42	
27	
52	

2) Given sum of angles of polygons find number of sides

Sum of angles	Number of sides
4140°	
2880°	
1620°	
2700°	

3) All inner angles of a 30 sided polygon are equal

i) How much is each outer angle

ii) How much is each inner angle

4) Is the sum of the angles of any polygon is 1500°

5) The sum of the angles of a polygon is 3060°. what is the sum of the angles of a polygon with one side more and for a polygon with one side less

6) All inner angles of a 15 sided polygon are equal

i) How much is each outer angle

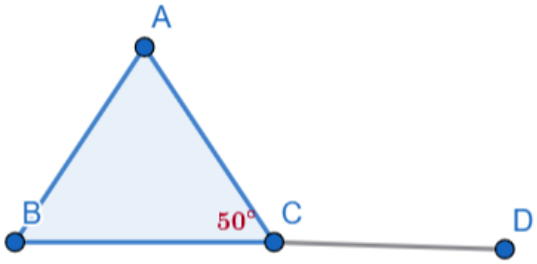
ii) How much is each inner angle

7) The sum of the angles of a polygon is 6300° how many sides does it have

8) In the figure BC extended toD at C. <ACB= 50 °

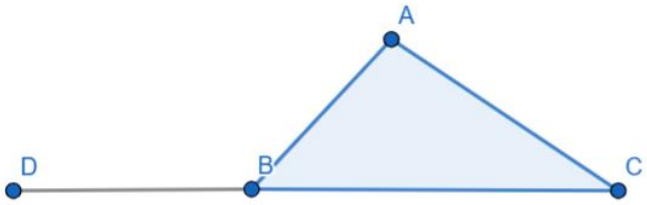
i)Find <ACD

i) Find <A+<B

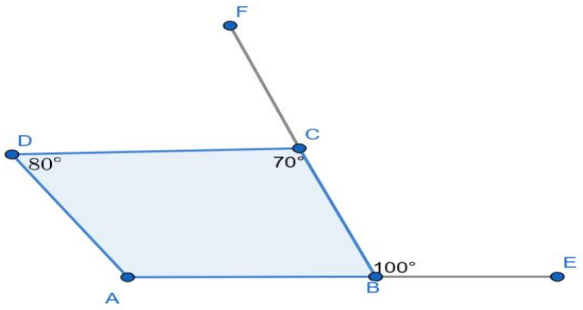


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9) In triangle ABC AB = AC and CB is extended toD at B. Outer angle at B is110°. Find all the angles of ∆ABC



10) In the figure, ABCD is a quadrilateral and AB is extended to E at B.BC is extended to F at C.



Find <BAD

Find <DCF

11) Complete the following table by finding the number of sides outer angle and inner angle of regular polygons

Number of sides	Outer angle	Inner angle
20		
	45°	
		144°
15		
	40°	
		150°
	20°	
		170°
	15°	

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